Q1)

Task1.java

```java
import java.util.Scanner;
public class Task1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your age: ");
        int age = sc.nextInt();
        if (age >= 18) {
            System.out.println("You can vote.");
        } else {
            System.out.println("You cannot vote.");
        }
    }
}
```

Output

```
Enter your age: 12
You cannot vote.

=== Code Execution Successful ===
```

Q2)

Task2.java

```java
import java.util.Scanner;

public class Task2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Principal: ");
        double p = sc.nextDouble();

        System.out.print("Enter Time: ");
        double t = sc.nextDouble();

        System.out.print("Enter Rate: ");
        double r = sc.nextDouble();

        double si = (p*t*r)/100;
        System.out.println("Simple Interest="+si);
    }
}
```

Output

```
Enter Principal: 2
Enter Time: 1
Enter Rate: 2
Simple Interest=0.04

=== Code Execution Successful ===
```

Q3)

Task.java

```java
import java.util.Scanner;

public class Task {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter breadth: ");
        float breadth = sc.nextFloat();
        System.out.print("Enter height: ");
        float height = sc.nextFloat();
        float areaTriangle = 0.5f *  breadth * height;

        System.out.println("Area of Triangle = " + areaTriangle);
        System.out.print("Enter side of cube: ");
        float side = sc.nextFloat();
        float volumeCube = side * side * side;
        System.out.println("Volume of Cube = " + volumeCube);

        System.out.print("Enter length, width, height: ");
        float l = sc.nextFloat();
        float w = sc.nextFloat();
        float h = sc.nextFloat();
        float volumeCuboid = l * w * h;
        System.out.println("Volume of Cuboid = " + volumeCuboid);
    }
}
```

Output

```
Enter breadth: 2
Enter height: 2
Area of Triangle = 2.0
Enter side of cube: 2
Volume of Cube = 8.0
Enter length, width, height: 3
2
3
Volume of Cuboid = 18.0

=== Code Execution Successful ===
```

Q4)

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        votingRule();
    }
    static void votingRule() {
    System.out.println("Enter your age: ");
    int age = sc.nextInt();

    String check = (age >= 18) ? "eligible to vote" : "not
        eligible to vote";

    System.out.println("You are " + check);
}
}
```

Output

```
Enter your age:
18
You are eligible to vote

=== Code Execution Successful ===
```

Q5)

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        sumProduct();
    }
    static void sumProduct() {
        System.out.print("Enter first number: ");
        int num1 = sc.nextInt();
        System.out.print("Enter second number: ");
        int num2 = sc.nextInt();
        int sum = num1 + num2;
        System.out.println("Sum of numbers " + num1 + " and " + num2 + " is: " + sum);
        int product = num1 * num2;
        System.out.println("Product of numbers " + num1 + " and " + num2 + " is: " + product);
    }
}
```

Output

```
Enter first number: 2
Enter second number: 3
Sum of numbers 2 and 3 is: 5
Product of numbers 2 and 3 is: 6

=== Code Execution Successful ===
```

Q6)

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        divisionTwo();
    }
    static void divisionTwo() {
        System.out.print("Enter first number: ");
        float num1 = sc.nextFloat();

        System.out.print("Enter second number: ");
        float num2 = sc.nextFloat();

        float sum = num1 + num2;
        float product = num1 * num2;
        float division = product / sum;

        System.out.println("Sum = " + sum);
        System.out.println("Product = " + product);
        System.out.println("Division (product/sum) = " + division);
    }
}
```

Output

```
Enter first number: 2
Enter second number: 3
Sum = 5.0
Product = 6.0
Division (product/sum) = 1.2

=== Code Execution Successful ===
```

Q7)

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        format();
    }
    static void format() {
        System.out.print("Enter your name: ");
        String name = sc.nextLine();
        System.out.print("Enter your roll number: ");
        int rollNumber = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter your field of interest: ");
        String interest = sc.nextLine();
        System.out.println("Hey, my name is " + name +
                " and my roll number is " + rollNumber +
                ". My field of interest is " + interest + ".");
    }
}
```

Output

```
Enter your name: Avigya
Enter your roll number: 7
Enter your field of interest: IT
Hey, my name is Avigya and my roll number is 7. My field of interest is
    IT.

=== Code Execution Successful ===
```

Q8)

AllInOne.java

```java
import java.util.Scanner;

public class AllInOne {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Side of square: ");
        double s = sc.nextDouble();
        System.out.println("Area = " + (s * s));
        System.out.println("Perimeter = " + (4 * s));
        System.out.print("\nPrincipal, Rate, Time: ");
        double p = sc.nextDouble();
        double r = sc.nextDouble();
        double t = sc.nextDouble();
        System.out.println("SI = " + (p * r * t / 100));
        System.out.print("\nBase, Height: ");
        double b = sc.nextDouble();
        double h = sc.nextDouble();
        System.out.println("Area = " + (0.5 * b * h));
        System.out.print("\nCube side: ");
        double c = sc.nextDouble();
        System.out.println("Volume = " + (c * c * c));
        System.out.print("\nLength, Width, Height: ");
        double l = sc.nextDouble();
        double w = sc.nextDouble();
        double ht = sc.nextDouble();
        System.out.println("Volume = " + (l * w * ht));
        sc.close();
    }
```

Output

```
Side of square: 4
Area = 16.0
Perimeter = 16.0

Principal, Rate, Time: 2
3
4
SI = 0.24

Base, Height: 1
2
Area = 1.0

Cube side: 3
Volume = 27.0

Length, Width, Height: 1

3
3
Volume = 9.0

=== Code Execution Successful ===
```

Q9)

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        areaRectangle();
    }
    static void areaRectangle() {
        System.out.println("enter length of rectangle: ");
        double length = sc.nextDouble();
        System.out.println("enter breadth of rectangle: ");
        double breadth = sc.nextDouble();
        int area = (int) Math.round(length * breadth);
        System.out.println("Area of rectangle with length " +
            length + " and breadth " + breadth + " is: " + area);
    }
}
```

Output

```
enter length of rectangle:

2
enter breadth of rectangle:
2
Area of rectangle with length 2.0 and breadth 2.0 is: 4

=== Code Execution Successful ===
```

Q10)

Week2Task.java

Share Run

```java
public class Week2Task {
    Scanner sc = new Scanner(System.in);
    void totalMarks() {
        System.out.println("Enter marks in Mathematics: ");
        Float mathematics = sc.nextFloat();
        System.out.println("Enter marks in Business: ");
        Float business = sc.nextFloat();
        System.out.println("Enter marks in Software: ");
        Float software = sc.nextFloat();
        System.out.println("Enter marks in Programming: ");
        Float programming = sc.nextFloat();
        Float total=mathematics + business + software+programming;
        Float percentage = (total / 4);
        String result;
        result = ((percentage >= 70) ? "First Class" :
                  (percentage >  59) ? "Upper Second Class" :
                  (percentage >  49) ? "Second Class" :
                  (percentage >  39) ? "Third Class" : "Fail");
        System.out.println("Total: "      + total);
        System.out.println("Percentage: " + percentage + "%");
        System.out.println("Result: "     + result);
    }
    public static void main(String[] args) {
        Week2Task check = new Week2Task();
        check.totalMarks();
        check.sc.close();
    }
}
```

Output

```
Enter marks in Mathematics:
70
Enter marks in Business:
90
Enter marks in Software:
75
Enter marks in Programming:
89
Total: 324.0
Percentage: 81.0%
Result: First Class

=== Code Execution Successful ===
```